WASTED FOOD MANAGEMENT APP

A PROJECT REPORT

Submitted by

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ABSTRACT

The sharp rise in the amount of food waste creates the need for charity in terms of donations. Food is mostly lost every day in numerous restaurants, parties, social gatherings, college canteens and many other social activities in the present scenario. By visiting each organization several times to reduce the problems with food waste, people donate food manually. Where there are platforms that have made attempts to help people donate food, the current framework offers a new internet- based application that provides all poor people/organizations with a forum for recycling surplus food. The device has seen to be an efficient way to donate items over the Internet to organizations, etc. It highlights the ability to prevent food waste. It offers details about the inspiration behind such an application, explaining the current mechanism of contributions and how the product works to benefit society. This framework would establish a shared communication platform for hotels/restaurants, charitiesandindividuals,wherecharitiesandindividualscancontactrestaurants that have remaining food directly and report generation. Showing how much food is given by which restaurant and giving reward points Food Donor, Food Receiver, Third Party Provider, Admin are the key modules in this framework where Food Donor can be any organization, entity or college that wants to donate food and build a new food donation request and Food Receiver can be any food-seeking charity corporation. A fresh food donation request will be produced on the site and a message will be sent to the third-party provider responsible for transporting food from the food donor to the food recipient until the request is approved.

Keywords-FoodDonation,Phone,Volunteer,Mobileapplication,Foodwastage reduction

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CHAPTER 1

INTRODUCTION

1.1 PROBLEM DEFINITION:

The Food Donation Portal Proposed Program is an internet-based application that provides all poor people/organizations with a forum for donating left over food.The product has seen to be a powerful way of donating items over the internet to organizations, etc. Food pollution is a troubling phenomenon in densely developed nations like India.

Too much food gets tossed out from weddings,canteens,pubs,social and family get-togethers and events.The scheme will bring them into practice instead of losing these items by donating them to different organizations such as orphanages,oldagehomes, etc. Some individuals and institutions want to contribute items to organizations in need.Many organizations often tend to inquire for different itemsthey need,suchas clothing, food grains, books, utensils, etc., but there is no source available that can meet their requirements. A mobile-based application has therefore been created in which individuals can donate food products according to their capacity, and the application often enables organizations toa pplythei rrequests.,i.e,items required by them, if any.

The food donation application serves as an interface between consumers searching for a channel to donate without wasting the surplus food.It encouragesus to donate the surplus food by notifying local users of the available food information. The requested users demand the notice.Depending on the priority,the machine allocates food products.

The proposed application eliminates food waste and thus satisfies other criteria, such as food goods from vulnerable organizations. The proposed scheme is currently aimed at eliminating the significant waste that normally exists in India, which is food.

1.2 EXISTINGAPPLICATIONS:

In the current system, if anybody has additional food in view of any potential or in their home, it would become squandered in view of the fact that it is extremely doubtful immediately. In the event of possessing tons of food, to convey to everybody. If they ought to find some refuge with the extra food Individuals or poor. They don't have time, or they don't care about it. So we have an application for funding for extra food for those in need or next to shelter**.**

1.3 NEED FOR THE SYSTEM:

We are reducing food waste by using the application in the proposed system. The role of relocating food is a massively fruitful social development that deals with food waste and the need for food. At that point, the administrator collects food from the donor via their close by specialist to the nearby shelters or poor people. The proposed application is android-based, generated using java and xml on Android Studio, involves site association and will provide a stage for contributors and searchers after they register effectively in the system. He/she should express something unique in the document on the unlikely chance that a client wants to offer something. This framework's UI will be simple and easy to use, and Android is based on the framework. As of now, we expect the major waste that usually exists in India to be eliminated and that is important. We are searching and hoping to refresh and refine a similar that will amount to the application's efficiency and usefulness, like books, fixed, clothes, etc.In any case, The application is limited to Android Mobile phones with Android OS and higher renderings. The application would still be profitable if contributors and searchers are similar to each other usage case graph shown above shows 3 entertainers: Donor, Recipient and Admin. Activities such as registration and signing into the system are carried out by the donor.

CHAPTER 2
LITERATURESURVEY

'Share Meal' is a Smartphone Android program that provides donors and seekers with a forum to donate and collect food once they have successfully logged into the system. system consists of three primary donor, volunteer and admin modules. The donor completes tasks such as registration/login and adds items to the donation request to be contributed and viewed. The recipient does tasks such as requesting items, displaying requested items and claiming donations. The manager will track the collection and upgrade it. The administrator and the donor will also see the position of the recipient. The donor-donated objects will be displayed to other users as a reminder in the donation tab and the message will be saved in the backend folder.

'Food donation portal', a paper published in 2015, briefly outlines the food donation practices and provides a forum connecting donors to NGOs. An concept to eliminate food waste, minimize food waste and improve the food donation network is introduced and an impact on society is made possible via this medium. The paper 'Beyond Food Sharing: Promoting the Elimination of Food Waste With ICTs', releasedin2016, maintaining food security is essential to tracking citizens' quality of life at various levels of society.

In the recent economic crisis, a significant number of people have been living in conditions of food hunger and poverty, especially in developing regions. Despite the knowledge and concern about the value of food and the elimination of food waste and food surplus management, the role of ICTs in this area is still uncertain and poorly reported.

CHAPTER 3

SYSTEMAN ALYSIS

3.1 SYSTEMREQUIREMENTS:

The following specifications were those required by the system for the software's successful implementation and functioning

3.1.1 REQUIREMENTS:

SOFTWARE REQUIREMENTS

- **OPERATINGSYSTEM**: ANDROIDOS
- **FRONT-END** : HTML,CSS,ANDBOOTSTRAP
- **BACK-END** : ANGULARJS,PHP,MYSQL

HARDWARE REQUIREMENTS

- **PROCESSOR/CPU**:

  An Intel i5 or AMD Ryzen 5 or higher is recommended. However, a lower-tier processor could suffice for basic usage.

- **MEMORY/RAM:**

  At least 8 GB of RAM is recommended. More RAM will help with performance, especially when running emulators and handling large projects.

- **DISK SPACE:**

  Android Studio itself requires a minimum of 4 GB of disk space, but you'll need more for your projects, system files, and emulators. A solid-state drive (SSD) is preferable for better performance.

- **OPERATING SYSTEM:**

  Android Studio is available for Windows, macOS, and Linux. Ensure you have a compatible OS version.

3.1.2 PACKAGES:

ANDROID STUDIO PACKAGES:

As of my knowledge cutoff in September 2021, the latest version of Android Studio was Android Studio 4.2. However, please note that there may have been newer releases since then. Android Studio is the official Integrated Development Environment (IDE) for Android app development, provided by Google. Regarding packages in Android Studio, there are various components and packages available depending on your requirements. Some commonly used packages include:

- **Android SDK Platform**:
  This package contains the Android API libraries and system images necessary for developing and testing Android apps.
- **Android Support Library**:
  This package provides backward compatibility features for older Android versions, allowing you to use newer APIs on older devices.
- **Google Play Services:**
  This package includes APIs for integrating various Google services, such as Google Maps, Firebase, Google Sign-In, and more, into your Android apps.
- **Android NDK(Native Development Kit):**
  This package allows you to develop and integrate native code (C/C++) into your Android apps.
- **Android Emulator**:
  This package provides virtual devices that allow you to test your Android apps on different screen sizes, device configurations, and Android versions.
- **Android Build Tools:**
  This package includes tools like ADB(Android Debug Bridge), Dex compiler, and other utilities necessary for building and deploying Android apps.

CHAPTER 4
SYSTEM DESIGN

4.1 OBJECT ORIENTED DESIGN:

Identifying the objects in a system would be what OO (Object Oriented) analysis and design have always been about identifying their relationships. Generating a design that could be transformed into executables utilizing object-oriented programming languages.

UML (Unified Modeling Language) is a standard language that uses, designs, constructing, and documenting software components. The Unified Modeling Language (UML) is an effective instrument enabling Object-Oriented Analysis and Design. The Object Management Group(OMG)created it, and in January 1997,the OMG proposed UML1.0 as a specification draught. It started as a way to capture the behavior of vast software and non-software systems and has since grown into such an international standard. UML is created to be process generic, indicating it could be used in a variety of circumstances.Itcanbeusedforaspectrumofuses.Businessanalysts,software architects, and developers are all using UML as a common language. It can be used to describe, specify, build, and document the system's business processes, including its structural and behavioral artifacts.

UML is a modeling language used to create software blueprints. Diagrams are categorized into two divisions, which are further divided into subcategories:

- **Structural Diagrams**
- **Behavioral Diagrams**

4.2 STRUCTURAL DIAGRAMS:

The static aspect of the system is portrayed by the structural diagrams. These static aspects are the components of a diagram that describes the main structure and are thus stable. Classes, interfaces, objects, components, and nodes are often used to represent them.

Some of the structural diagrams are:

- Class diagram
- Object diagram
- Component diagram
- Deployment diagram

4.3 BEHAVIORAL DIAGRAMS:

Behavioral diagrams illustrate a system's complex nature. The changing/moving parts of a system are usually known for the dynamic aspect. The following five types of behavioral diagrams are supported in UML:

4.3.1 Use case diagram

4.3.2 Sequence diagram

4.3.3 Collaboration diagram

4.3.4 State chart diagram

4.3.5 Activity diagram

4.4 USECASEDIAGRAM:

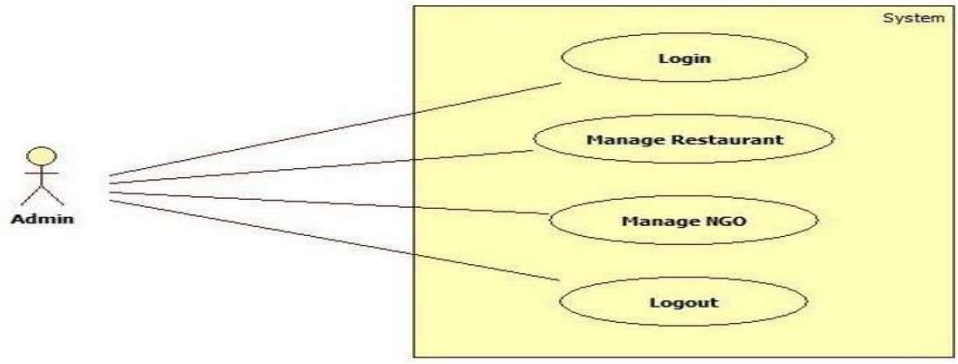


Fig. Use Case Diagram of Admin

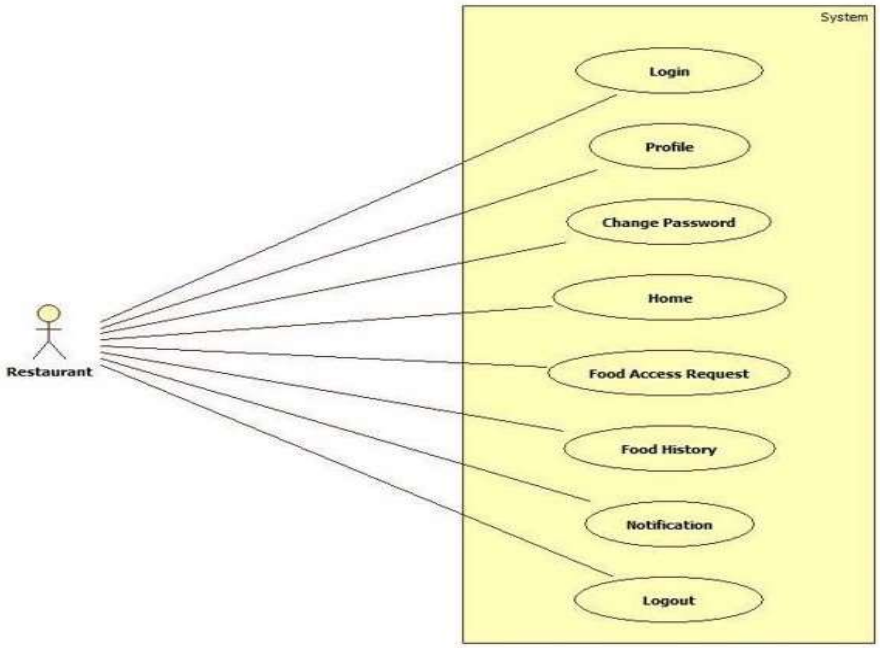


Fig. Use Case Diagram of Restaurant

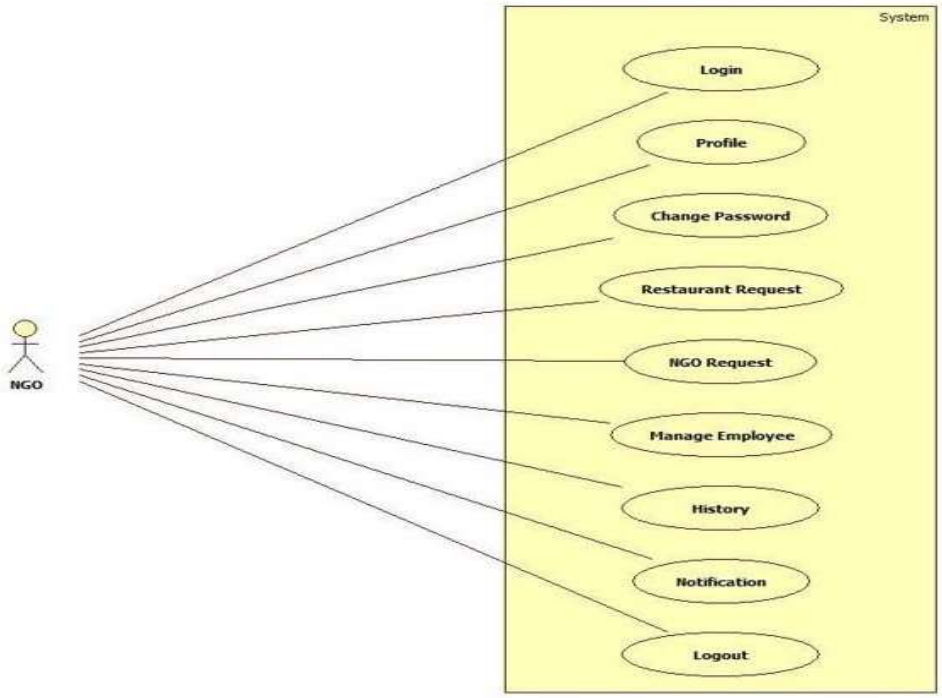
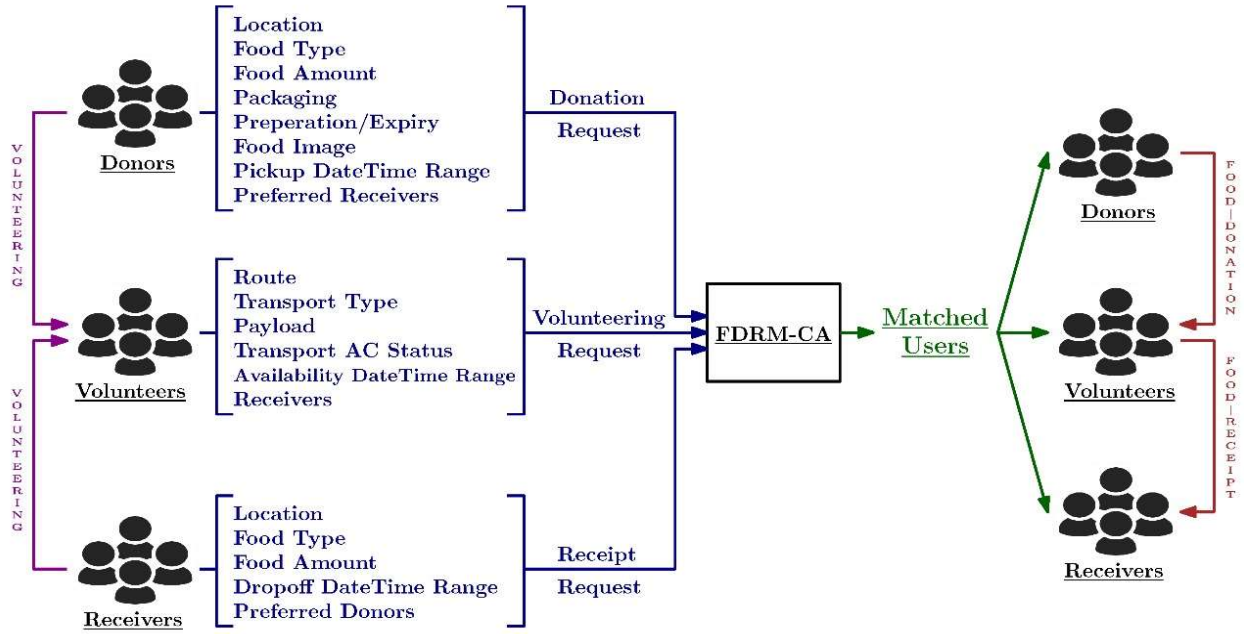
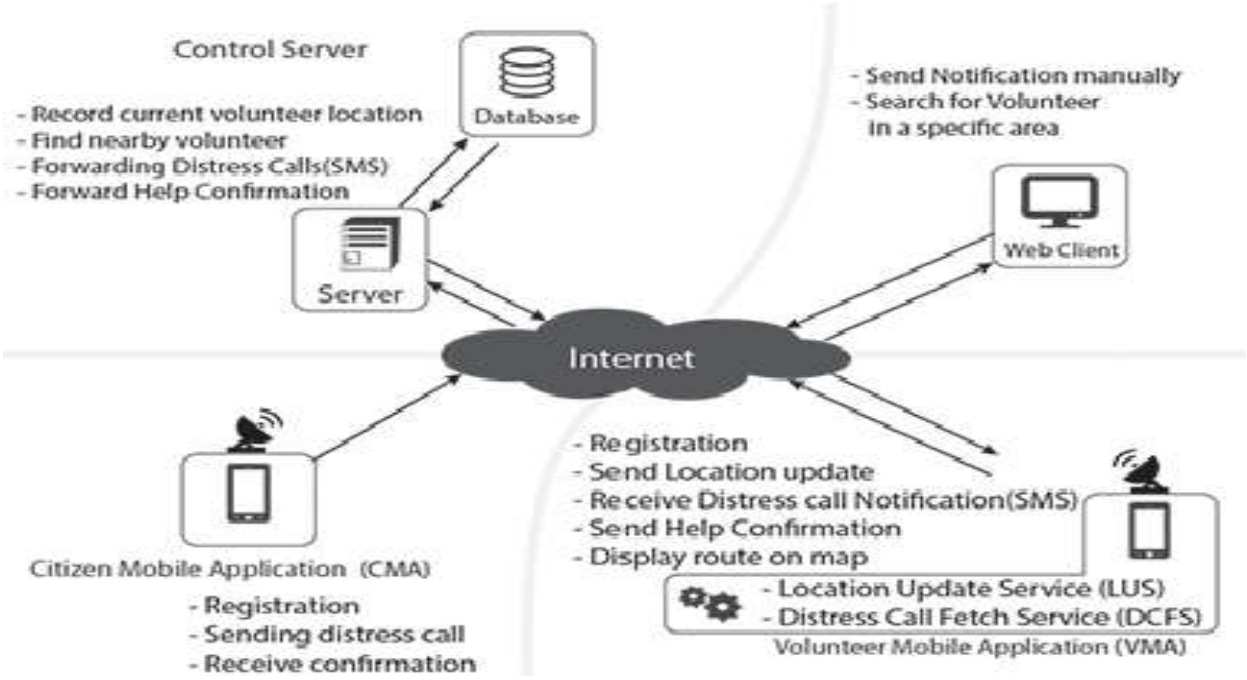


Fig. Use Case Diagram of NGO

4.5 ARCHITECTURE DIAGRAM:





CHAPTER 5

SYSTEM IMPLEMENTATION

5.1 ALGORITHMS AND PROPOSED TECHNIQUES:

5.1.1 FOOD DONATION

- Considering the anomalies in the existing system computerization of the whole activity is being suggested after initial analysis.
- The android application is developed using Android Studio with JAVA as a programming language.
- Proposed system is accessed by one entity namely, User.
- Admin need to login with their valid login credentials first in order to access the android application.
- After successful login, admin can access all the modules and perform/manage each task accurately.
- Admin can perform task such as can manage restaurants and NGO's by adding them and update the list.
- Restaurant can login and update their profiles. They can also view the accepted food list which is yet to pick up. Restaurant can add access food details. They can also accept request from NGO.
- Restaurant can also view the accepted, pending and previous today's Access Food list which are accepted by NGO.
- They will be getting notifications. NGO can login and update their profile by providing details.
- They can view and accept the restaurants request and also food details. They can accept and assign an employee for food pick up.
- In the time of food shortage NGO can also raise request to the restaurants. And after request been accepted by restaurant, they can assign an employee for delivery purpose. They will get notifications.

5.1.2 FOOD DONATION ALGORITHM

STEP1: Start the application

STEP2: Register by filling the necessary details.

STEP3: Select the Option of donor or volunteer accordingly.

STEP4: If you wish to donate go to donation page by clicking on Donor button.

STEP 5: Then click on create donation.

STEP6: Fill up the necessary details and click Submit.

STEP7: If you wish to see donation request then click on Request for Food.

STEP8: If you are in need of food then go to request food option which is on volunteer page.

STEP9: Fill up your requirement and click on Request button.

STEP10: If you wish to see available food list by donors click on Available Food.

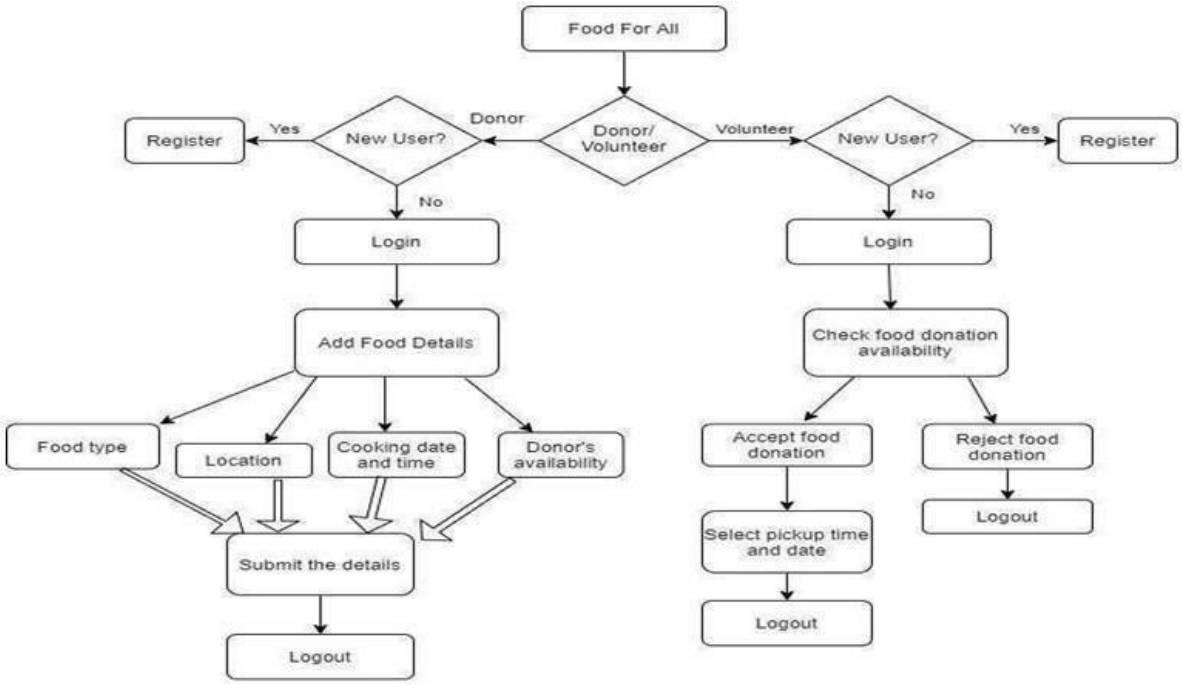


Fig1.3FOOD DONATION APP FLOW CHART

CHAPTER 6
FUNCTIONALITIES

1. LOGIN&REGISTRATION:

For both the guest and the agent, this stage involves login and enlistment. By preserving separate documents for each customer, the subtleties of the client are kept classified. The expert can only see the subtleties of the enlisted guest concurrently.

2. ADMINISTRATOR MODULE:

In administrator module, the over seer keeps up the specialist subtleties just as the donator subtleties. The executive gathers the food from the specialist. The overseer gives the shelter subtleties straight forwardly to the donator.

3. DONATOR MODULE:

In the donor module, the donor allocates the food waste to the shelter. In this way we can reduce food waste issue, the donor gives the solicitation to the donator. The suggested application is android-based, generated using java and xml on Android Studio, requires web association and will offer contributors and searchers a stage after they register effectively in the system.

He/sheshouldexpresssomethinguniqueinthedocumentontheunlikelychancethata client wants to offer something. This note would appear as a notification to various customers in the Gifts column. This message would be stored in the database on the backend. The shelters who want to guarantee the gifts can reply to the contributor and reach him/her when a note is received

This framework's UI will be simple and easy to use, and Android is based on the framework. As of now, we expect the major waste that usually exists in India to be eliminated and that is important. We are searching and hoping to refresh and refine a similarthatwillamounttotheapplication'sefficiencyandusefulness,likebooks,fixed, clothes, etc.

In any event, the program is limited to Android Smart phones with Android OS and higher renderings. If contributors and searchers are placed next to each other, be optimistic.

4. RECIPIENT MODULE:

In Agent module, the Receiver keep up the halfway house subtleties. It can likewise keep up the donator subtleties. The Receiver give the solicitation to the administrator forgatherthefoodfromthedonator.Aftergatherthefoodthespecialistgivesthealarm message for the donator.

CHAPTER 7
PERFORMANCE METRICS

When evaluating the performance of a food donation app, there are several key metrics that can be considered. These metrics can help measure the effectiveness, efficiency, and impact of the app. Here are some performance metrics for a food donation app:

USER ENGAGEMENT:

- Number of app downloads and installations.
- Daily, weekly, and monthly active users (DAU, WAU, and MAU).
- User retention rate.
- Average session duration.
- Number of user interactions (such as donations made, requests fulfilled, messages sent).

DONATIONS:

- Total number of food donations made through the app.
- Total weight or volume of food donated.
- Average donation size (in terms of weight, items, or meals).
- Distribution of donated food by category (fresh produce, canned goods, etc.).
- Donation frequency per user (how often users make donations).

REQUESTS FULFILLED:

- Total number of food requests fulfilled through the app.
- Total weight or volume of food distributed to fulfill requests.
- Average fulfillment time for requests.
- Distribution of fulfilled requests by category (urgent, long-term, specific dietary needs, etc.).
- Percentage of requests fulfilled successfully.

USER FEEDBACK AND RATINGS:

- App store ratings and reviews.
- User satisfaction surveys or feedback forms.
- Net Promoter Score (NPS) or similar measures of user advocacy.

OPERATIONAL EFFICIENCY:

- Response time to user inquiries or support requests.
- Number of technical issues, bugs, or crashes encountered.
- App performance metrics (e.g., load time, app size).
- Server uptime and reliability.
- Time and effort required for on boarding new users or organizations.

IMPACT AND OUTREACH:

- Number of individuals or families supported by the app.
- Number of partner organizations (food banks, shelters, etc.) involved.
- Geographical coverage or reach of the app.
- Reduction in food waste (in terms of estimated food diverted from waste).

CHAPTER 8

CONCLUSION AND FUTURE WORK

8.1 CONCLUSION:

The proposed application would eliminate food waste and also fulfill various prerequisites, such as clothes, books, utensils, and so on, of penniless associations. There is a great deal of food waste that exists every day at eateries and bistros, as described above in the illustration. Instead of discarding junk equivalents (which typically is the situation It can be used very effectively to take care of the destitute. Similarly, because the pickup is arranged by the corporation, the cafés/bistros need not think about it. Both the eateries/bistros (reducing carbon impression and waste) and the penniless would benefit In future jobs, there was no standard food data system on food packages that offers the consumer the data on both the name of the food package, just as its expiry date. The required change will be to get the name of the food from the generic identification of the object and read the expiry date using OCR appliances.

Be it as it might, the degree of convenience of using this option is just marginally more remarkable than using the manual alternative that of fills the food data. In order to provide detailed details, a few organizations have initiated preliminaries using QR code on their food packages. In any event, certain hurdles do need to be passed in order to make it into a standard. Yet this application provides a realistic and powerful arrangement for the interim.

8.2 FUTURE WORK:

Our study has investigated the problem of food waste that has many serious side effects economically and socially. However, the wastage of the food can be prevented or at least decreased using political rules and technology. Mobile application technology is beneficial for food waste management. The app aims to encourage better food management.. This work is an initial step towards designing a better system to reduce daily food waste. In future, this app could be enhanced more by adding the following features: Extending our app to have many types of donating users either from organizations such as restaurants, or a family or a single user Adding the location (GPS) facility to our apps. The donating user should specify the location of the shared food. Adding the time and date of each meal shared by users Making the app supports multiple platforms (cross-platform app).

APPENDIX

APPENDIXA-SOURCECODE

A. CODE FOR ADMIN

AdminActivity.java

```
packagecom.kirandroid.sharemeal.activities.administrator;

import android.os.Bundle;
importandroid.view.MenuItem;
import android.view.View;
import android.widget.Button;
importandroid.widget.EditText;
importandroid.widget.TextView;
import android.widget.Toast;
importandroidx.annotation.NonNull;
importandroidx.appcompat.app.AppCompatActivity;
importcom.kirandroid.sharemeal.R;
importcom.kirandroid.sharemeal.modals.Volunteer;
import com.google.firebase.database.DataSnapshot;
import com.google.firebase.database.DatabaseError;
importcom.google.firebase.database.DatabaseReference;
import com.google.firebase.database.FirebaseDatabase;
import com.google.firebase.database.Query;
import com.google.firebase.database.ValueEventListener;

publicclassAdminActivityextendsAppCompatActivity{

   TextViewtxtName,txtPhone,txtEmail,txtCount,txtNormal,txtHigh;
   Button btnSearch;
   EditTextedtPhone;
   Stringname,email,phone;
   int count;
   DatabaseReferencedatabaseReference;
   @Override
   protectedvoidonCreate(BundlesavedInstanceState){
      super.onCreate(savedInstanceState);
      setContentView(R.layout.activity_admin);
```

```
        if (getSupportActionBar() != null) {
            getSupportActionBar().setDisplayHomeAsUpEnabled(true);
            getSupportActionBar().setTitle("Administrator");
        }
        txtName = (TextView) findViewById(R.id.admintxtName);
        txtPhone = (TextView) findViewById(R.id.admintxtPhone);
        txtEmail = (TextView) findViewById(R.id.admintxtEmail);
        txtCount = (TextView) findViewById(R.id.admintxtCount);
        txtNormal=(TextView)findViewById(R.id.admintxtNormal);
        txtHigh = (TextView) findViewById(R.id.admintxtHigh);
        edtPhone = (EditText) findViewById(R.id.adminedtPhone);
        btnSearch = (Button) findViewById(R.id.adminbtnSearch);
        databaseReference =
FirebaseDatabase.getInstance().getReference().child("Volunteers");
        databaseReference.keepSynced(true);
        btnSearch.setOnClickListener(newView.OnClickListener(){
            @Override
            publicvoidonClick(Viewv){

                StringvolPhone=edtPhone.getText().toString().trim(); if

                (volPhone.isEmpty()) {
                    edtPhone.setError("PleaseenterVolunteerPhoneNumber");
                } else if (volPhone.length() < 10) {
                    edtPhone.setError("PhoneNumberisInvalid!");

                }else {
                    //Toast.makeText(AdminActivity.this,""+volphone,
Toast.LENGTH_SHORT).show();

                    Query query =
databaseReference.orderByChild("phone").equalTo(volPhone);

                    query.addListenerForSingleValueEvent(newValueEventListener(){
                        @Override
                        publicvoidonDataChange(@NonNullDataSnapshotdataSnapshot)
{
                            if(dataSnapshot.exists()){
```

```
                    for(DataSnapshotdataSnapshot1:dataSnapshot.getChildren())
{

                        Volunteervolunteer=
dataSnapshot1.getValue(Volunteer.class);
                        if(volunteer!=null){
                            name = volunteer.getName();
                            email = volunteer.getEmail();
                            phone=volunteer.getPhone();
                            count = volunteer.getCount();
                        }else {
                            Toast.makeText(AdminActivity.this,"NoDataFound!",
Toast.LENGTH_SHORT).show();
                        }

                    }
                    txtName.setText(name);
                    txtPhone.setText(phone);
                    txtEmail.setText(email);
                    txtCount.setText(String.valueOf(count));

                    if (count > 10) {
                        txtHigh.setVisibility(View.VISIBLE);
                    }else {
                        txtNormal.setVisibility(View.VISIBLE);
                    }
                }else {
                    Toast.makeText(AdminActivity.this,"NoDataFound!",
Toast.LENGTH_SHORT).show();
                }
            }

            @Override
            publicvoidonCancelled(@NonNullDatabaseErrordatabaseError){
                Toast.makeText(AdminActivity.this, "" +
databaseError.getMessage(),Toast.LENGTH_SHORT).show();
            }
        });
    }
  }
});
```

```
    }

    @Override
    publicbooleanonOptionsItemSelected(@NonNullMenuItemitem){ int
       id = item.getItemId();
       if(id==android.R.id.home){

          onBackPressed();

       }
       returnsuper.onOptionsItemSelected(item);
    }
}
```

B. CODE FOR DONOR:

DonorActivity.java

```
packagecom.kirandroid.sharemeal.activities.donor;
importandroid.annotation.SuppressLint;

import android.app.DatePickerDialog;

import android.content.Context;

importandroid.content.Intent;

importandroid.net.ConnectivityManager;

import android.net.NetworkInfo;

import android.net.Uri;

importandroid.os.Bundle;

importandroid.view.MenuItem;

import android.view.View;

import android.widget.AdapterView;

importandroid.widget.ArrayAdapter;

import android.widget.Button;import

android.widget.DatePicker; import

android.widget.EditText; import

android.widget.Spinner;
```

```
importandroid.widget.Toast;
importandroidx.annotation.NonNull;
importandroidx.appcompat.app.AppCompatActivity;
importcom.kirandroid.sharemeal.R;
importcom.kirandroid.sharemeal.modals.Happy;
importcom.google.firebase.database.DatabaseReference;
import com.google.firebase.database.FirebaseDatabase;
importjava.util.Calendar;
publicclassDonorCelebrateActivityextendsAppCompatActivityimplements
AdapterView.OnItemSelectedListener {
   EditTextdon_name,don_email,don_phone,don_money,don_add,don_date; String
   name, email, phone, money, recip, key, address, date, occasion;
   Buttonsubmit;
   Stringadmin_1,admin_2,admin_3,msg;
   DatabaseReference myref;
   booleanconnected=false;
   @Override
   protectedvoidonCreate(BundlesavedInstanceState){
      super.onCreate(savedInstanceState);
      setContentView(R.layout.activity_happy);
      don_name=(EditText)findViewById(R.id.celebdonname);
      finalSpinnerspinneroccasion=findViewById(R.id.celeboccasionspin);
      don_phone=(EditText)findViewById(R.id.celebdonphone);
      don_email = (EditText) findViewById(R.id.celebdonemail);
      don_add = (EditText) findViewById(R.id.celebdonaddress);
      don_date = (EditText) findViewById(R.id.celebdate);
      final Spinner spinner = findViewById(R.id.celebspin);
      don_money=(EditText)findViewById(R.id.celebdonmoney);
      submit = (Button) findViewById(R.id.celebsubmit);
```

```
myref = FirebaseDatabase.getInstance().getReference().child("Happy_Moments");
        admin_1 ="9381384234";
        admin_2 ="8639796138";
        admin_3 ="6303149161";

        don_date.setOnClickListener(newView.OnClickListener(){
           @Override
           public void onClick(View v) {
              Calendarc=Calendar.getInstance();
              int mYear = c.get(Calendar.YEAR);
              intmMonth= c.get(Calendar.MONTH);
              intmDay=c.get(Calendar.DAY_OF_MONTH);
              DatePickerDialog datePickerDialog = new
DatePickerDialog(DonorCelebrateActivity.this,new
DatePickerDialog.OnDateSetListener() {
    //DatePicker.getDatePicker().setMaxDate(System.currentTimeMillis());
                  }
              },mYear,mMonth, mDay);

    datePickerDialog.getDatePicker().setMinDate(System.currentTimeMillis()-
    1000);
              datePickerDialog.show();
           }
        });
        final ArrayAdapter<CharSequence> occasionadapter =
    ArrayAdapter.createFromResource(DonorCelebrateActivity.this,
    R.array.occasion,android.R.layout.simple_spinner_dropdown_item);
    occasionadapter.setDropDownViewResource(android.R.layout.simple_spinner_dr
    opdown_item);
        spinneroccasion.setAdapter(occasionadapter);
        spinneroccasion.setOnItemSelectedListener(DonorCelebrateActivity.this);
```

```
        final ArrayAdapter<CharSequence> adapter =
ArrayAdapter.createFromResource(this,R.array.recipient,
android.R.layout.simple_spinner_dropdown_item);
adapter.setDropDownViewResource(android.R.layout.simple_spinner_dropdown
_item);
        spinner.setAdapter(adapter);
        spinner.setOnItemSelectedListener(DonorCelebrateActivity.this);
        submit.setOnClickListener(newView.OnClickListener(){
           @Override
           publicvoidonClick(Viewv){
              name =don_name.getText().toString().trim();
              occasion=spinneroccasion.getSelectedItem().toString(); email
              = don_email.getText().toString().trim();
              phone = don_phone.getText().toString().trim();
              money=don_money.getText().toString().trim();
              recip = spinner.getSelectedItem().toString();
              address = don_add.getText().toString().trim();
              date = don_date.getText().toString().trim();
              StringemailPattern="[a-zA-Z0-9._-]+@[a-z]+\\.+[a-z]+";
              if (name.isEmpty()) {
                 don_name.setError("PleaseenterName!");
              } else if (occasion.equals("Select Occasion")) {
                 Toast.makeText(DonorCelebrateActivity.this,"Pleaseselect
Occasion",Toast.LENGTH_SHORT).show();
              }elseif(phone.isEmpty()){
```

```
            don_phone.setError("PleaseenterPhoneNumber");
        } else if (phone.length() < 10) {
            don_phone.setError("PhoneNumberisInvalid");
        } else if (email.isEmpty()) {
            don_email.setError("PleaseenterEmailID");
        } else if (!email.matches(emailPattern)) {
            don_email.setError("EmailisInvalid");
        } else if (address.isEmpty()) {
            don_add.setError("PleaseenterAddress");
        } else if (date.isEmpty()) {
            Toast.makeText(DonorCelebrateActivity.this,"PleasechooseDateof
Celebration",Toast.LENGTH_SHORT).show();
        } else if (recip.equals("Select Recipient")) {
            Toast.makeText(DonorCelebrateActivity.this,"Pleaseselect
Recipient",Toast.LENGTH_SHORT).show();
        } else if (money.isEmpty()) {
            don_money.setError("PleaseenterMoney");
        }elseif(money.equals("0")){
            Toast.makeText(DonorCelebrateActivity.this,"Pleaseenteran
amount grater than 0", Toast.LENGTH_SHORT).show();
        }else {
            ConnectivityManagerconnectivityManager=(ConnectivityManager)
getSystemService(Context.CONNECTIVITY_SERVICE);
            if
(connectivityManager.getNetworkInfo(ConnectivityManager.TYPE_MOBILE).g
etState() == NetworkInfo.State.CONNECTED ||
connectivityManager.getNetworkInfo(ConnectivityManager.TYPE_WIFI).getSta
```

```
te()==NetworkInfo.State.CONNECTED){
                    //weareconnectedtoanetwork
                    Happyhappy_1_modal=newHappy(name,occasion,email, phone,
money, recip, address, date);
                    myref.push().setValue(happy_1_modal);
                    msg="DearAdministrator,\n"+name+"isreadytodonatea
sumof";
                    Stringres =msg +"Rs."+money +"to "+recip +"for"+
occasion+"occasion";
                    Stringre=res+",PleasecollectMoneyat"+ address;
                    admin_1 ="9381384234";
                    admin_2 ="8639796138";
                    admin_3 ="6303149161";
                    UrisendSmsTo=Uri.parse("smsto:"+admin_1+";"+admin_2+ ";" +
admin_3);
                    Intent intent = new Intent(
                          Intent.ACTION_SENDTO,sendSmsTo);
                    intent.putExtra("sms_body",re);
                    startActivity(intent);
                    connected=true;
                }else{
                    connected = false;
                    Toast.makeText(DonorCelebrateActivity.this,"Internet
Unavailable",Toast.LENGTH_LONG).show();
                }
//myref=
FirebaseDatabase.getInstance().getReference().child("Happy_Moments").child(ke y);
              }
            }
          });
      }
```

```
    @Override
    publicvoidonItemSelected(AdapterView<?>parent,Viewview,intposition, long
id) {
    }
    @Override
    publicvoidonNothingSelected(AdapterView<?>parent) {

    }
    @Override
    publicbooleanonOptionsItemSelected(@NonNullMenuItemitem){ int
        id = item.getItemId();
        if(id==android.R.id.home){
            onBackPressed();
        }
        returnsuper.onOptionsItemSelected(item);
    }
}
```

C. MAIN ACTIVITY

MainActivity.java

```
packagecom.kirandroid.sharemeal.activities.general;
import android.content.Intent;
import android.graphics.Color;
import android.os.Bundle;
importandroid.view.MenuItem;
import android.view.View;
import android.widget.Button;
importandroid.widget.TextView;
importandroidx.annotation.NonNull;
importandroidx.appcompat.app.ActionBarDrawerToggle;
import androidx.appcompat.app.AppCompatActivity;
import androidx.appcompat.widget.Toolbar;
importandroidx.core.view.GravityCompat;
importandroidx.drawerlayout.widget.DrawerLayout;

importcom.kirandroid.sharemeal.R;
import com.kirandroid.sharemeal.activities.donor.DonorCelebrateActivity;
import com.kirandroid.sharemeal.activities.donor.FoodDetailsActivity;

import com.kirandroid.sharemeal.activities.recipients.RecipientLoginActivity;
import com.kirandroid.sharemeal.activities.volunteer.VolunteerActivity;
importcom.kirandroid.sharemeal.activities.volunteer.VolunteerLoginActivity;
import com.kirandroid.sharemeal.adapter.BannerAdapter;
importcom.kirandroid.sharemeal.modals.Banners;
importcom.kirandroid.sharemeal.recyclerView.RecyclerViewGallery;
```

```
import com.kirandroid.sharemeal.utilities.ConstantValues;
importcom.kirandroid.sharemeal.utilities.MyAppPrefsManager;
import com.google.android.gms.ads.AdRequest;
import com.google.android.gms.ads.AdSize;
importcom.google.android.gms.ads.AdView;
importcom.google.android.gms.tasks.OnCompleteListener;
import com.google.android.gms.tasks.Task;
importcom.google.android.material.navigation.NavigationView;
import com.google.firebase.auth.FirebaseAuth;
importcom.google.firebase.messaging.FirebaseMessaging;
import com.google.firebase.storage.FirebaseStorage;
import com.google.firebase.storage.StorageReference;
import com.smarteist. IndicatorView.animation.type.IndicatorAnimatioType;
importcom.smarteist.autoimageslider.SliderAnimations;
import com.smarteist.autoimageslider.SliderView;
importjava.util.ArrayList;
import java.util.List;
publicclassMainActivityextendsAppCompatActivityimplements
NavigationView.OnNavigationItemSelectedListener {
   publicButtonb1,b2;
   privateFirebaseAuthfirebaseAuth;
   private DrawerLayout drawer;
   public TextView t, textView;
   MyAppPrefsManagermyAppPrefsManager;
   booleandoubleBackToExitPressedOnce=false;
   SliderViewsliderImage;
   List<Banners>modelList=newArrayList<>();
   privateAdViewmAdView;
   privatestaticfinalStringTAG="MAIN_ACTIVITY";
```

```
//InterstitialAdinterstitialAd=null;
 protectedvoidonCreate(BundlesavedInstanceState){
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);
   //MobileAds.initialize(this,getString(R.string.admob_app_id));
    Toolbartoolbar=findViewById(R.id.toolbar);
    toolbar.setTitle("Meal 4 All");
    drawer = findViewById(R.id.drawer_layout);
    sliderImage=findViewById(R.id.slider_image);
    myAppPrefsManager = new MyAppPrefsManager(this);
    NavigationViewnavigationView=findViewById(R.id.nav_view);
    navigationView.setNavigationItemSelectedListener(this);
    ActionBarDrawerToggletoggle=newActionBarDrawerToggle(this,drawer, toolbar,
    R.string.navigation_drawer_open,R.string.navigation_drawer_close);
    drawer.addDrawerListener(toggle);
    toggle.syncState();
    firebaseAuth=FirebaseAuth.getInstance();
    listAllFiles();
    mAdView = new AdView(MainActivity.this);
    mAdView.setAdSize(AdSize.BANNER);
    mAdView.setAdUnitId(getString(R.string.banner_home_footer));
    mAdView=(AdView)findViewById(R.id.adView)
    AdRequestadRequest=newAdRequest.Builder().build();
    //interstitialAd=newInterstitialAd(MainActivity.this);
   //interstitialAd.(getString(R.string.interstitial_full_screen));
   //AdRequestadRequest1=newAdRequest.Builder().build();
   //interstitialAd.loadAd(adRequest1);
  /*mAdView.loadAd(adRequest);
```

```
mAdView.setAdListener(newAdListener(){ @Override
        publicvoidonAdLoaded(){
        }
        @Override
        publicvoidonAdClosed(){
        }
        @Override
        publicvoidonAdFailedToLoad(interrorCode){
        }
        @Override
        publicvoidonAdLeftApplication() {
        }
        @Override
        publicvoidonAdOpened(){
           super.onAdOpened();
        }
      });*/
      FirebaseMessaging.getInstance().subscribeToTopic("/topics/donateFood")
            .addOnCompleteListener(newOnCompleteListener<Void>(){
               @Override
               publicvoidonComplete(@NonNullTask<Void>task){
                  String msg = "getString(R.string.msg_subscribed)";
                  if(!task.isSuccessful()){
                      msg="getString(R.string.msg_subscribe_failed)";
                  }
               }
            });

      findViewById(R.id.volun).setOnClickListener(newView.OnClickListener(){
         @Override
         publicvoidonClick(Viewview){ try
            {
               if(ConstantValues.IS_USER_LOGGED_IN=
 myAppPrefsManager.isUserLoggedIn()) {
```

```
                Intentintent=newIntent(getBaseContext(),VolunteerActivity.class);
                intent.setFlags(Intent.FLAG_ACTIVITY_CLEAR_TASK |
Intent.FLAG_ACTIVITY_CLEAR_TOP);
                startActivity(intent);
              }else{
                Intentintent=newIntent(getBaseContext(),
VolunteerLoginActivity.class);
                intent.setFlags(Intent.FLAG_ACTIVITY_CLEAR_TASK|
Intent.FLAG_ACTIVITY_CLEAR_TOP);
                startActivity(intent);
              }

            }catch(Exceptione){
              e.printStackTrace();
            }
          }
        });
        findViewById(R.id.rest).setOnClickListener(newView.OnClickListener(){
          @Override
          publicvoidonClick(Viewview){
            Intenti=newIntent(MainActivity.this,FoodDetailsActivity.class);
            i.setFlags(Intent.FLAG_ACTIVITY_CLEAR_TASK |
Intent.FLAG_ACTIVITY_CLEAR_TOP);
            startActivity(i);
          }
        });
      }
      publicvoidlistAllFiles(){
        FirebaseStoragestorage=FirebaseStorage.getInstance();
        StorageReferencestorageRef=storage.getReference().child("bannerImages");
        StorageReference storageRef1 = storage.getReference();
        storageRef.listAll();
```

```
modelList.clear();
                for(StorageReferenceitem:listResult.getItems()){
                   //AlltheitemsunderlistRef.

                   //[STARTdownload_via_url]

 storageRef1.child(item.getPath()).getDownloadUrl().addOnSuccessListener(uri->{

                       modelList.add(new Banners(item.getName(), uri.toString()));
                       BannerAdapteradapter=newBannerAdapter(MainActivity.this,
 modelList);
                       sliderImage.setSliderAdapter(adapter);

 sliderImage.setIndicatorAnimation(IndicatorAnimationType.SWAP);
 sliderImage.setAutoCycleDirection(SliderView.AUTO_CYCLE_DIRECTION_BAC
 K_AND_FORTH);
                       sliderImage.setIndicatorRadius(5);
                       sliderImage.setIndicatorSelectedColor(Color.WHITE);
                       sliderImage.setIndicatorUnselectedColor(Color.GRAY);
                       sliderImage.startAutoCycle();
                       sliderImage.setOnIndicatorClickListener(position-
                            >sliderImage.setCurrentPagePosition(position));
                   }).addOnFailureListener(exception->{
                   });
               }
})
            .addOnFailureListener(e->{
            });
    }
```

```
    @Override
    publicbooleanonNavigationItemSelected(@NonNullMenuItemitem){
        switch (item.getItemId()) {
            caseR.id.nav_about:
                Intentb=newIntent(MainActivity.this,IssuesActivity.class);
                b.setFlags(Intent.FLAG_ACTIVITY_CLEAR_TASK |
Intent.FLAG_ACTIVITY_CLEAR_TOP);
                startActivity(b);
                break;
            caseR.id.nav_gallery:
                Intentg=newIntent(MainActivity.this,RecyclerViewGallery.class);
                g.setFlags(Intent.FLAG_ACTIVITY_CLEAR_TASK |
Intent.FLAG_ACTIVITY_CLEAR_TOP);
                startActivity(g);
                break;
            caseR.id.nav_org:
                Intentintent=newIntent(MainActivity.this,RecipientLoginActivity.class);
                intent.setFlags(Intent.FLAG_ACTIVITY_CLEAR_TASK |
Intent.FLAG_ACTIVITY_CLEAR_TOP);
                startActivity(intent);
                break;
            caseR.id.nav_happy:
                Intentintent1=newIntent(MainActivity.this,
DonorCelebrateActivity.class);
                intent1.setFlags(Intent.FLAG_ACTIVITY_CLEAR_TASK|
Intent.FLAG_ACTIVITY_CLEAR_TOP);
                startActivity(intent1);
                break;
            caseR.id.nav_doncount:
```

```
            Intentintent3=newIntent(MainActivity.this,DonationCount.class);
            intent3.setFlags(Intent.FLAG_ACTIVITY_CLEAR_TASK |
  Intent.FLAG_ACTIVITY_CLEAR_TOP);
            startActivity(intent3);
            break;
          caseR.id.nav_nearby:
            Intentintent4=newIntent(MainActivity.this,NearbyNGOActivity.class);
            intent4.setFlags(Intent.FLAG_ACTIVITY_CLEAR_TASK|
  Intent.FLAG_ACTIVITY_CLEAR_TOP);
            startActivity(intent4);
            break;
        }
        drawer.closeDrawer(GravityCompat.START);
        return true;
    }
    @Override
    publicvoidonBackPressed(){
    }
    @Override
    publicvoidonPause(){
        if(mAdView!=null){
            mAdView.pause();
        }
        super.onPause();
    }
    @Override
    publicvoidonResume(){
        super.onResume();
        if(mAdView!=null){
            mAdView.resume();
}
```

APPENDIX B

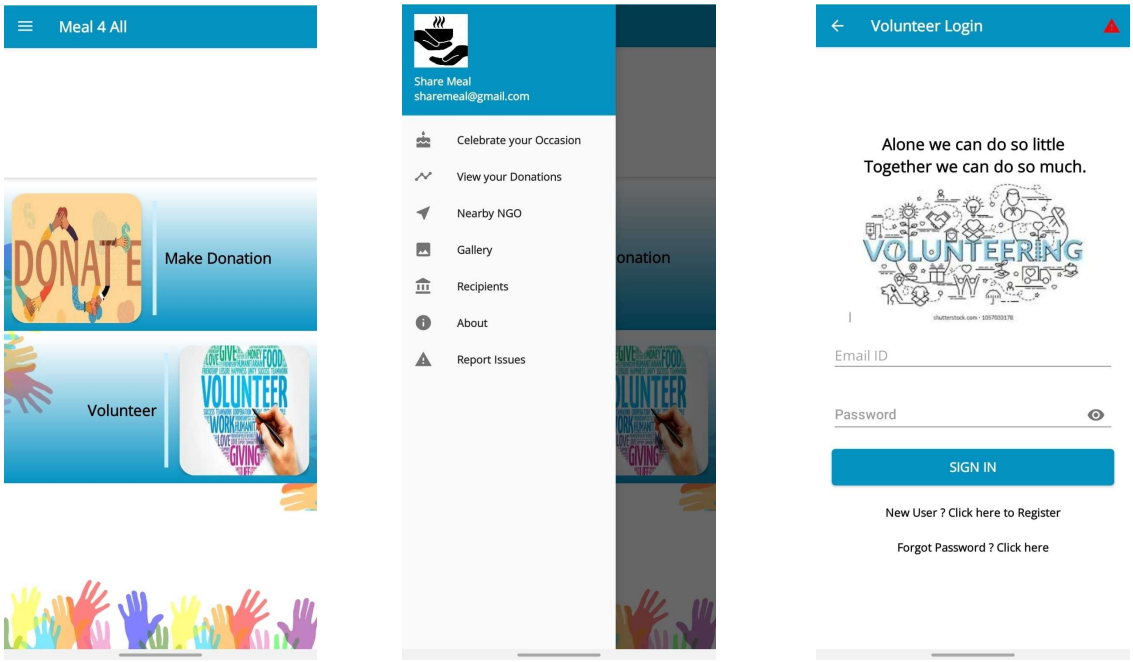


Fig1.4Output snap shot of main page and volunteer login

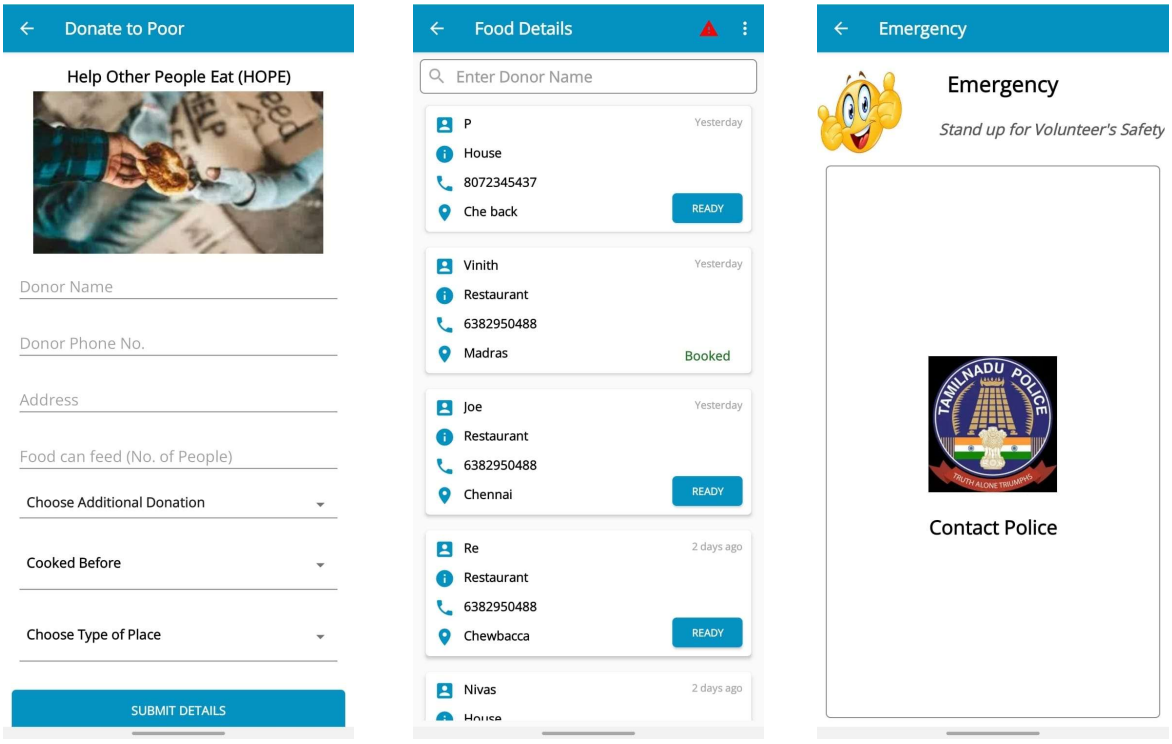


Fig1.5Output snap shot of donor login

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